Supplementary Information

for: Landfill methane recovery and use under deep uncertainty: a satellite-constrained Monte Carlo decision model for the Norte III landfill, Buenos Aires

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S1. Identifiability: variance shares, contraction and dependence

Eq. (12) of the manuscript gives the 2022 collection efficiency from the satellite emission, the oxidation fraction and the reported capture. It holds to 1e-16 in the sampled draws. Because the log-odds of capture is a sum of three terms, its variance divides into shares: in the prior (draws consistent with $\eta \le 0.95$) the emission carries 97% of the variance of $\ln[(1-\eta)/\eta]$ (standard deviation 0.41), the reported capture 3.4% and oxidation −0.0%; in the lognormal posterior the shares are 89%, 7% and 3.9% (standard deviation 0.27). Table S1 gives the prior-to-posterior contraction of the inputs that enter generation; contraction is 1 − IQR(posterior)/IQR(prior), and KS is the Kolmogorov–Smirnov distance between the two marginals.

Input	Prior, median (P5–P95)	Posterior, median (P5–P95)	Contraction	KS
Waste acceptance A (t/d)	16,500 (14,791–18,209)	16,561 (14,862–18,243)	1%	0.026
Methane potential L0 (Nm³ CH_4/t)	105 (67–150)	113 (79–151)	13%	0.138
Steady-state fraction f_ss	0.61 (0.40–0.84)	0.65 (0.46–0.85)	13%	0.130
Methane fraction	0.50 (0.44–0.55)	0.50 (0.44–0.55)	−0%	0.010
Oxidation fraction	0.18 (0.09–0.30)	0.17 (0.09–0.29)	4%	0.033
Collected biogas 2022 (Nm³/h)	13,049 (12,332–13,700)	13,047 (12,329–13,702)	−1%	0.006
Generation G22 (Nm³ CH_4/h)	43,527 (24,087–72,266)	49,642 (34,300–70,876)	26%	0.238
Efficiency η 2022	0.15 (0.09–0.27)	0.13 (0.09–0.19)	43%	0.231

Table S1. Prior-to-posterior contraction (prior: draws consistent with $\eta \le 0.95$; posterior: lognormal).

Posterior correlations (Pearson, of logarithms for the generation terms) are: ln L0 with ln f_ss −0.33 (prior 0.00), ln G22 with η22 −0.94, ln E22 with η22 −0.92, ln L0 with η22 −0.55 and ln f_ss with η22 −0.50. The negative correlation between L0 and f_ss is the ridge along which generation is constant; the satellite constraint selects the ridge, not a point on it.

S2. Photovoltaic coating against conventional photovoltaics

On the covered area, conventional PV has a positive NPV in 4.0% of draws and a median LCOE of US$119 (84 to 169) per MWh, above the merchant value of electricity because of the landfill foundation premium, the capital premium and the discount rate. The paint, with deliberately wide priors (efficiency 2 to 10%, installed cost US$5 to 150 per m², life 2 to 20 years), has a median LCOE of US$211 (60 to 605) per MWh and is cheaper in 23% of draws. Setting the levelised cost of the paint equal to that of conventional PV gives the break-even efficiency

$$\eta^{*} = \frac{c_{pt}\ \kappa\ \left[CRF\left(r, L_{pt}\right) + o_{pt}\right]}{LCO\, E_{\mathrm{conv}}\ H_{poa}\ PR} \tag{S1}$$

which has a median of 9.7% (2.9% to 24.9%) and exceeds 10% in 48% of draws (Fig. S2a). The 23% figure is a property of unvalidated priors; the transferable result is the η^{*} surface. Verified against the simulation: setting the paint efficiency to η^{*} gives LCOE parity to numerical precision (automated test).

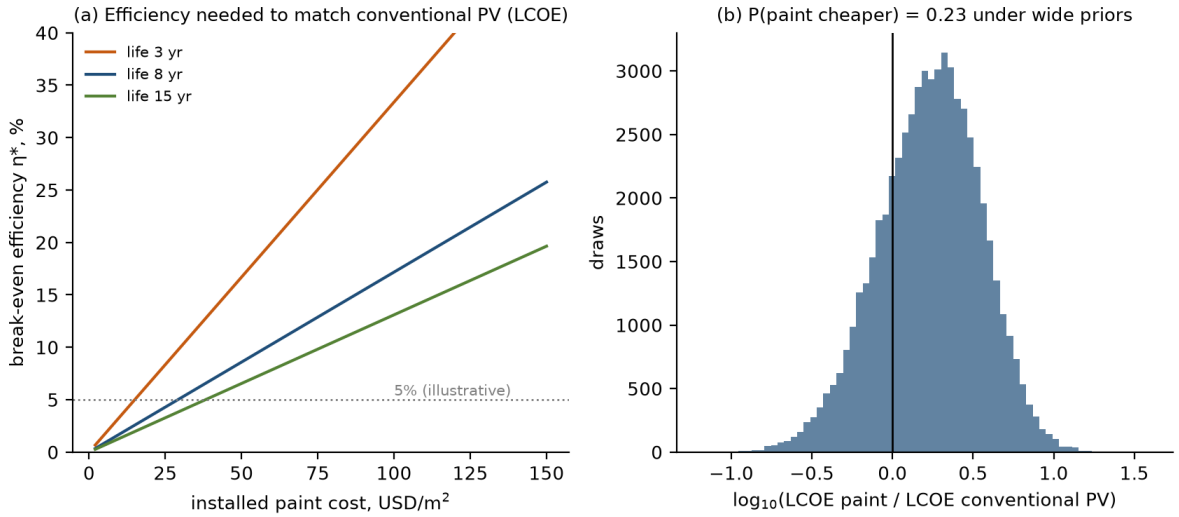


Fig. S2. (a) Break-even efficiency of PV paint for LCOE parity with conventional PV against installed cost, for three lifetimes (other inputs at posterior medians). (b) Distribution of the LCOE ratio.

S3. Input distributions

Table S2 lists the sampled inputs. Distributions are triangular (minimum, mode, maximum), uniform (minimum, maximum) or lognormal (σ, mean 1). Tags: R, operator-reported or published programme value; L, literature or standard value; B, dated market or regulatory benchmark (Table 2 of the manuscript); A, assumed by the author and requiring a site-specific or dated quotation before use in a decision. Inputs marked * enter only the robustness variants. The sensitivities of Sections 3.8 and 3.9 of the manuscript (alternative generation priors, satellite scaling, engine slip 1 to 6%, marginal grid factor 0.35 to 0.65 with mode 0.45, counterfactual flare destruction 0.80 to 0.98, displacement 0.3 to 1.0) are tag A and are defined in NorteIII_review.py.

Input	Description	Distribution	Tag
Qcoll22	collected biogas around 2022 (satellite period), Nm3/h; 13,100 peak in 2013, 13,800 in 2025	tri(12,000, 13,100, 14,000)	A
Awaste	waste to Norte III, t/day (~18,000 reported)	tri(14,000, 16,500, 19,000)	R
L0	CH4 generation potential, Nm3 CH4 per t waste (inventory 100, CAA 170)	tri(50, 100, 170)	L
fss	fraction of steady-state generation reached (age/decay/lag)	tri(0.3, 0.6, 0.95)	A
xCH4	CH4 fraction of landfill gas	tri(0.42, 0.5, 0.57)	A
ox	cover methane oxidation fraction	tri(0.05, 0.15, 0.35)	L
g_t	generation 2025-26 relative to 2022	tri(0.95, 1.05, 1.15)	A
Qcoll	collected biogas 2025-26, Nm3/h (13,800 in 2025; 16,000 Apr 2026)	tri(13,000, 14,800, 16,500)	R
P_el	installed biogas power, MW (23 in 2020; 33 planned 2026)	tri(23, 28, 33)	R
CF_el	engine capacity factor	tri(0.7, 0.85, 0.93)	A
eta_el	engine electrical efficiency	tri(0.36, 0.4, 0.43)	L
d_f	flare/engine CH4 destruction efficiency	tri(0.95, 0.98, 0.995)	L
dEta1	S1 absolute gain in collection efficiency (of generation)	tri(0, 0.02, 0.06)	A
capex_s1	S1 capex, MUSD	tri(0.5, 1.5, 4)	A
opex_s1	S1 opex, MUSD/yr	tri(0.15, 0.4, 1)	A
f_hot	flux fraction passing covered area	tri(0.1, 0.25, 0.5)	A
h_e	hot-spot flux enrichment over mean	tri(1.5, 3, 6)	A
A_act	actively emitting area, ha	tri(100, 200, 400)	A
eps_conv	conventional cover: fraction of arriving CH4 blocked	tri(0.3, 0.6, 0.85)	A
eps_coat	experimental coating: fraction blocked	tri(0.1, 0.45, 0.85)	A
s_c	share of blocked CH4 captured by extraction	tri(0.05, 0.35, 0.75)	A
delta_ox	fractional loss of oxidation under cover	tri(0.2, 0.6, 1)	A

Input	Description	Distribution	Tag
rho_r	oxidation of redirected gas relative to ox	uni(0, 1)	A
c_conv	conventional cover cost, kUSD/ha	tri(80, 150, 300)	A
c_coat	experimental coating cost, kUSD/ha	tri(100, 250, 700)	A
om_cov	cover O&M, fraction of capex per yr	tri(0.015, 0.03, 0.05)	A
life_conv	conventional cover life, yr	tri(15, 25, 30)	A
life_coat	coating life, yr	tri(4, 8, 15)	A
r	real USD discount rate (EMBI 2025: 561 to >1,200 bps; 569 bps Dec 2025)	tri(0.08, 0.12, 0.18)	B
f_imp	imported share of equipment capex	tri(0.5, 0.7, 0.9)	A
duty	import duty on capital goods (0% BK regime; 12.6% Decree 513/2025; 20% Mercosur CET)	tri(0, 0.126, 0.2)	B
logist	freight, logistics and FX cost on imports, fraction	tri(0.03, 0.08, 0.15)	A
m_p	realised-price multiplier (tariff/FX risk), mean 1	logn(0.25)	A
avail_firm	fraction of surplus gas contractually firm	tri(0.6, 0.85, 1)	A
uptime	plant availability	tri(0.8, 0.9, 0.95)	A
rec	CH4 recovery in upgrading	tri(0.9, 0.95, 0.985)	L
tail_ox	fraction of tail-gas CH4 slip oxidised	uni(0, 0.95)	A
fug_plant	plant fugitive CH4, fraction of CH4 in	tri(0.002, 0.006, 0.012)	L
capex_up	upgrader capex, USD per Nm3/h raw capacity	tri(1,800, 3,000, 5,000)	A
e_up	upgrading electricity, kWh/Nm3 raw	tri(0.2, 0.28, 0.4)	L
c_el	electricity purchase cost, USD/MWh (wholesale monomic ~87-91 USD/MWh 2024-25 plus network)	tri(70, 95, 130)	B
om_up	upgrader O&M, fraction of capex per yr	tri(0.03, 0.04, 0.06)	A
lab_up	labour/other, MUSD/yr	tri(0.15, 0.25, 0.45)	A
capex_gv	GNC compression capex, USD per Nm3/h product	tri(400, 800, 1,500)	A
capex_gf	GNC fixed capex, MUSD	tri(0.3, 0.8, 2)	A
p_gnc	bio-GNC net plant-gate price, USD/Nm3 (delivered station gas ~0.15, retail ~0.40-0.45, 2025)	tri(0.13, 0.19, 0.4)	B
p_pipe	pipeline net plant-gate price, USD/Nm3 (PIST 2.94-2.98 USD/MMBtu, Dec 2025)	tri(0.07, 0.1, 0.16)	B
D_gnc	bio-GNC anchor demand, Nm3/h	tri(300, 1,000, 3,000)	A
dist_km	distance to injection point, km	tri(2, 8, 20)	A
c_km	pipeline spur cost, MUSD/km	tri(0.3, 0.6, 1.2)	A
c_inj	injection/metering station, MUSD	tri(0.5, 1, 2)	A
capex_pw	gas-engine power capex, USD/kW (RenovAr landfill-gas ref 1.3 MUSD/MW)	tri(1,100, 1,300, 1,700)	R
p_el	electricity sale price, merchant/MAT, USD/MWh (marginal cost 69, monomic ~87-91, 2024-25)	tri(60, 75, 95)	B
p_el_k*	electricity sale price, RenovAr-type contract, USD/MWh (landfill-gas awards 128-130, 2017)	tri(110, 125, 135)	B
cf_pw	new engines capacity factor	tri(0.8, 0.88, 0.93)	A
eta_pw	new engines electrical efficiency	tri(0.36, 0.4, 0.43)	L
om_pw	engine O&M, fraction of capex per yr	tri(0.03, 0.045, 0.07)	A
var_loss	flow-variability loss without buffer	uni(0.05, 0.25)	A
zeta	fraction of variability loss removed by buffer	uni(0.5, 0.9)	A
capex_buf	buffer capex, MUSD	tri(0.3, 0.8, 2)	A
leak_buf	buffer CH4 leakage, fraction of throughput	tri(0.002, 0.005, 0.01)	A
f_pv	share of upgrader electricity from on-site PV	uni(0.1, 0.5)	A
dEta_ad	adaptive-control gain in collection efficiency	tri(0, 0.01, 0.03)	A
capex_ad	adaptive control capex, MUSD	tri(0.2, 0.5, 1.2)	A
opex_ad	adaptive control opex, MUSD/yr	tri(0.05, 0.1, 0.25)	A
kwp_ha	conventional PV, kWp per ha	tri(500, 700, 900)	A
capex_pv	PV capex, USD/kW (RenovAr ref 850)	tri(550, 750, 1,100)	R
prem_pv	landfill foundation premium for PV	tri(1.1, 1.3, 1.6)	A
yield_pv	PV yield, kWh/kWp/yr	tri(1,350, 1,500, 1,650)	A
om_pv	PV O&M, fraction of capex per yr	tri(0.01, 0.015, 0.025)	A
life_pv	PV life, yr	tri(20, 25, 30)	A
p_pv	PV electricity value, merchant, USD/MWh (2024-25 wholesale cost)	tri(55, 70, 90)	B
eta_pt	PV-paint module efficiency (unvalidated)	tri(0.02, 0.05, 0.1)	A
c_pt	PV-paint installed cost, USD/m2 (unvalidated; wide on purpose)	tri(5, 30, 150)	A
life_pt	PV-paint life, yr (unvalidated; wide on purpose)	tri(2, 6, 20)	A

Input	Description	Distribution	Tag
pr_pt	PV-paint performance ratio	tri(0.65, 0.78, 0.85)	A
om_pt	PV-paint O&M, fraction of capex per yr	tri(0.01, 0.02, 0.04)	A
H_poa	plane-of-array irradiation, kWh/m2/yr	tri(1,550, 1,650, 1,750)	A
ef_grid	grid emission factor, tCO2/MWh	tri(0.28, 0.31, 0.36)	A
cp*	carbon-credit price if eligible, USD/tCO2e (VCM ~5.5, landfill gas ~9.5 USD, 2025)	tri(2, 6, 12)	B
p_elig*	probability that added capture is eligible for credits	uni(0.2, 0.8)	A
dv*	debt share of financing	tri(0, 0.4, 0.6)	A
r_e*	real cost of equity	tri(0.12, 0.16, 0.22)	A
r_d*	real cost of debt	tri(0.07, 0.09, 0.12)	A
k_dec*	first-order decay rate, 1/yr (IPCC 2006, temperate wet MSW)	tri(0.05, 0.09, 0.15)	L
age*	years of deposition before 2026	tri(20, 28, 32)	A
t_close*	years until deposition stops, from 2026	tri(3, 8, 15)	A

Table S2. Input distributions.

S4. Balances and automated tests

Scenario	Emitted	Oxidised	Destroyed	Exported	Biogenic C	Exported energy	Electricity	Total CO_2e
S0	231.5	47.7	44.8	0.0	69.2	0.0	0.0	6.25
S1	224.6	46.2	53.2	0.0	74.5	0.0	0.0	6.06
S2	215.3	39.1	68.9	0.0	82.0	0.0	0.0	5.81
S4	215.3	39.1	68.9	0.0	82.0	0.0	0.0	5.81
S6	215.4	39.1	64.0	5.0	78.3	69.4	6.3	5.80
S6b	224.7	46.2	48.2	5.0	70.9	69.2	6.2	6.06
S7	211.9	38.3	68.3	5.0	80.9	69.4	4.3	5.71

Table S3. Scenario balances for the median draw: methane emitted, oxidized, destroyed and exported (kt CH_4 yr^{-1}); biogenic carbon released as CO_2 (kt C yr^{-1}); biomethane energy exported and grid electricity used (GWh yr^{-1}); total emissions (Mt CO_2e yr^{-1}, GWP100). Residuals of all balances are below 10^{-9} of generation.

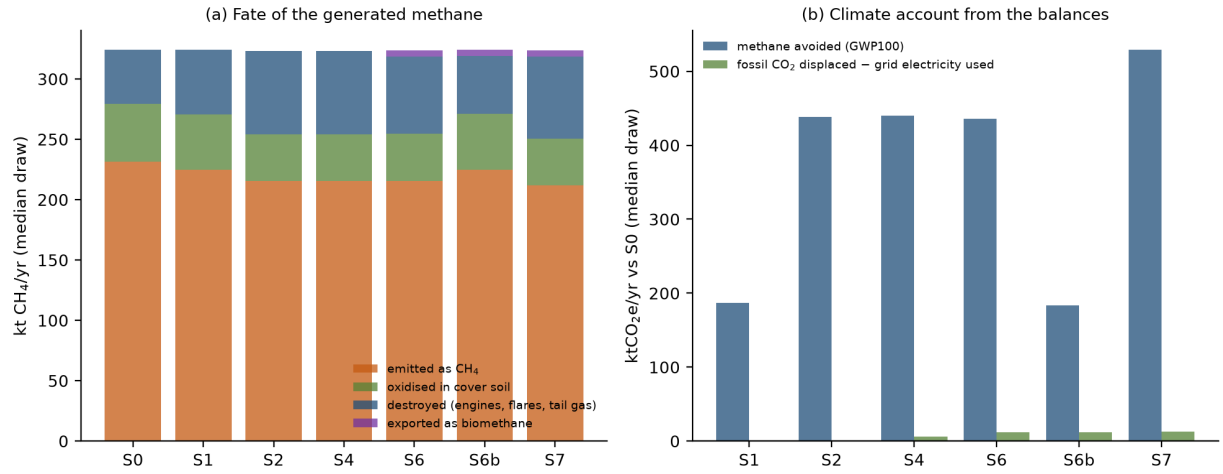


Fig. S3. (a) Fate of the generated methane by scenario (median draw). (b) Climate account from the balances: methane avoided (GWP100) and fossil CO_2 displaced net of grid electricity used, relative to S0.

The automated test suite (python NorteIII_MC.py --test) contains 59 checks, all of which pass: unit conversions (methane mass, carbon content, annuity and capital recovery factors); Latin-hypercube moments and the induced rank correlation; the white paper's screening arithmetic; closure of the cover balance; the analytic break-even s_c^*; two limiting cases of the cover; methane and carbon closure of seven scenario balances; methane and energy

closure of four upgrader accounts; equality of balance-based and incremental climate benefits for four scenarios; a hand-calculated upgrading case; an explicit cash-flow benchmark of the annuity NPV; unprofitability at zero gas prices; LCOE parity at η^*; the calibration window; physical consistency of the posterior; determinism; the steady-state limit of the time-resolved NPV and the analytic slow-decay limit of the first-order-decay model; the closed-form identity of the collection efficiency and the additive identity of its log-odds; the uninformative limit of the observation model; the reproduction of the annuity NPV by the engine counterfactual in the constant-supply and fast-decay limits; the capital-and-O&M floor under full curtailment; the zero of the engines' climate benefit at the break-even slip; the consistency of the conditional and unconditional probabilities; and the Monte Carlo standard error.

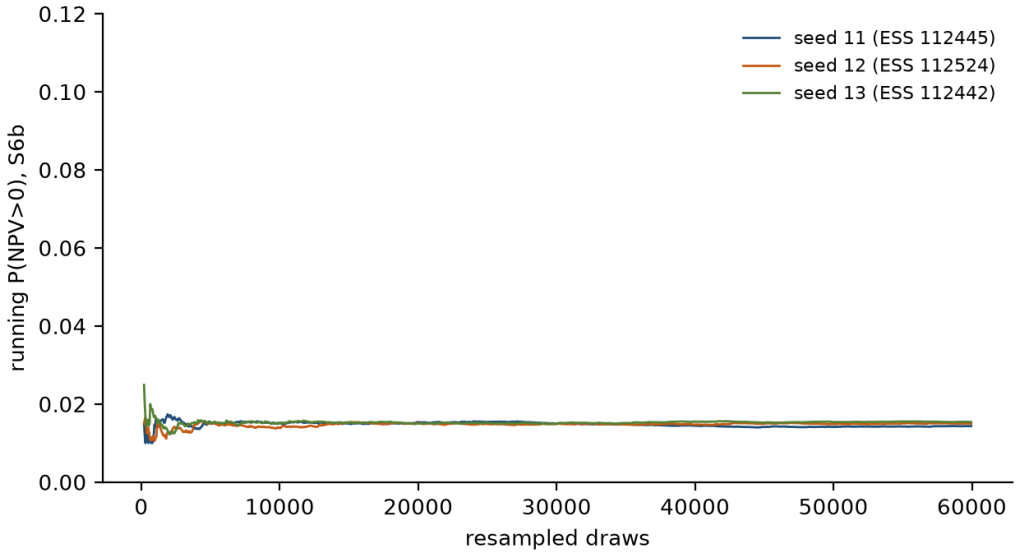


Fig. S1. Running probability of positive NPV for S6b across three independent seeds (resampled lognormal posterior; ESS in the legend).

S5. Scorecard for the research hypotheses

Hypothesis	Result
H1: cover plus pressure-managed extraction lowers total emissions	Supported conditionally: net reduction in 92% of draws if $s_c > s_c^*$ (median 0.14); net benefit is 64% of gross capture.
H2: PV coatings beat conventional PV only above measurable thresholds	Paint cheaper in 23% of draws under wide, unvalidated priors; break-even efficiency 9.7% at median inputs.
H3: biomethane from surplus gas has higher risk-adjusted NPV than alternatives	Not supported: beats additional engines in 0.02% of draws; break-even US$0.46 per Nm³ (US$14 per MMBtu).
H4: buffering offsets its leakage, energy and capital costs	Not supported: positive in 0.03% of draws; median ΔNPV −1.18 MUSD.
H5: the integrated configuration has lower lifecycle CO_2e per tonne of waste	Supported but small: S7 lower in 99.6% of draws; median reduction 8.3%, from 1.04 to 0.95 tCO_2e per tonne.

Table S4. Outcome of the five hypotheses of the project white paper (lognormal posterior).